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Direct Integration of Polycrystalline Diamond With 3C-SiC for Enhanced Thermal Management in GaN HEMTs: Impact of Grain Structure and Interface Engineering

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The direct integration of polycrystalline diamond (PCD) with semiconductors is crucial for enhancing heat dissipation in high-power electronics. However, achieving low surface roughness (<1 nm) remains challenging. In this study, the direct bonding of PCD to 3C-SiC for GaN high-electron-mobility transistors (HEMTs) on a 2-inch PCD wafer is demonstrated using an advanced bonding technique. The PCD wafer (surface roughness: 2.48 nm) is bonded at room temperature, forming a 7 nm-thick amorphous layer, which transformed into a 13 nm-thick polycrystalline SiC layer after annealing at 1100 °C without cracks or separations. Thermal analysis revealed higher thermal conductivity of PCD's growth surface than single-crystal diamond (SCD). However, the thermal resistance (R_{TH}) of GaN HEMTs on PCD is 27% higher than on SCD, attributed to phonon scattering from smaller grain sizes on the nucleation surface. Removing the fine-grained nucleation layer can enhance heat dissipation. This successful direct bonding of PCD with 3C-SiC overcomes key integration challenges, enabling improved thermal transport and high-power device reliability. To fully utilize PCD's thermal advantages, grain size optimization and interface engineering are essential to reduce phonon scattering, improve thermal transport efficiency, and maximize device performance for next-generation high-power electronics.

1. Introduction

Wide bandgap semiconductors such as GaN, SiC, and β-Ga_2O_3 offer high breakdown voltage and low specific on-resistance, enabling the development of compact and efficient power and RF electronic devices.[1] These materials are promising for next-generation applications, including radar systems, satellite communications, 5G stations, renewable energy, electric vehicles, energy storage, and consumer electronics.[1,2] However, Joule heating in these devices generates localized hot spots, leading to elevated temperatures that degrade the devices' performance and reliability.[3,4] As a result, thermal management has become a critical challenge in the electrical-thermal co-design of modern semiconductor devices. Heat dissipation challenges are becoming even more severe with increasing device miniaturization, integration, and power density.[5,6] Advanced thermal management solutions, including high thermal conductivity materials, innovative packaging techniques, and enhanced cooling technologies, are being actively researched to address these issues.[7–12] Effective heat spreading in localized hotspots requires a focused approach to minimizing thermal resistance in semiconductor heterostructures, particularly at substrates and interfaces.

GaN-based electronics have become a leading technology for high-power and high-frequency applications.[13] However, thermal limitations remain a key bottleneck in realizing maximum output power, particularly for GaN transistors operating near their electronic limit.[14] The hot spots in AlGaN/GaN high-electron-mobility transistors (HEMTs) exhibit exceptionally high heat flux, often an order of magnitude higher than the Sun's surface.[15] This underscores the critical role of heat spreading in thermal design to prevent excessive peak temperatures and enhance device reliability. Addressing these thermal bottlenecks is crucial for unlocking the full potential of GaN HEMTs in high-performance power and RF applications. Moreover, the median lifetime of GaN devices declines by half with every 10 °C increase in channel temperature, making thermal management a

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key determinant of device longevity.[16] As a result, today's GaN electronics are fundamentally constrained by thermal considerations, driving ongoing research into advanced cooling solutions, high thermal conductivity substrates, and innovative packaging techniques to overcome these limitations.

Diamond has the highest thermal conductivity among all natural materials, making it an ideal heat-spreading substrate for high-power and high-frequency semiconductor devices.[17] Due to its exceptional thermal properties, diamond has been extensively studied for the efficient cooling of GaN HEMTs, which are critical for applications in power electronics, RF systems, and high-frequency communication technologies.[18–21] However, integrating diamonds with GaN HEMTs remains a significant challenge, requiring advanced processing techniques and specialized materials to ensure minimal thermal resistance and high-quality diamond substrates. The GaN-on-diamond heterostructure has been developed using two primary approaches: diamond deposition and bonding. In the diamond deposition method, Si is removed from the GaN-on-Si substrate, and the diamond is directly deposited onto the exposed GaN surface using a dielectric material transition layer such as SiN_x or AlN.[20,22,23] The Si is also removed from the GaN-on-Si substrate in the diamond bonding. Still, instead of direct deposition, the GaN layer is bonded to a diamond substrate using an adhesion layer, typically composed of metal, amorphous Si, or other intermediate materials.[24–26]

The diamond deposition method has successfully produced 4-inch GaN-on-diamond wafers.[27] However, the low crystal quality of the diamond deposited on the SiN dielectric layer further degrades thermal conductivity,[28] limiting its effectiveness in high-performance thermal management. The diamond bonding method offers greater flexibility in optimizing interface properties to reduce thermal boundary resistance. However, more than 2-inch bonding and electrical insulator interface remain challenges for practical applications. Recent advancements have enabled the direct bonding of GaN and single-crystal diamond (SCD) at room temperature, demonstrating high thermal stability.[29] A 1-inch GaN HEMT on an SCD wafer was successfully fabricated using wafer bonding technology, where the bonding interface exhibited high thermal conductance.[30] GaN HEMTs integrated with SCD achieved more than twice the heat dissipation performance compared to those on SiC wafers, highlighting the superior thermal management capabilities of diamond-based integration.[30,31] This breakthrough paves the way for next-generation high-power GaN devices, where efficient heat dissipation is critical for device reliability and performance.

SCD is synthesized using Chemical Vapor Deposition (CVD) and High-Pressure High-Temperature (HPHT) methods.[32] Despite its promising properties, SCD is limited to a maximum size of ≈2 inches, restricting scalability for larger semiconductor wafers.[33] Additionally, SCDs are prohibitively expensive to produce, posing a challenge for large-scale commercialization. By contrast, polycrystalline diamond (PCD) deposited on Si has been successfully developed and scaled up to 8 inches.[34] Unlike SCDs, PCD offers a more cost-effective solution while maintaining excellent thermal conductivities. Its ability to be mass-produced at a controlled cost makes PCD the most promising heat dissipation material for GaN-on-diamond applications.

PCD is typically deposited on substrates such as silicon, where it nucleates from nano-sized diamond seeds. As these seeds coalesce into a continuous film, the diamond grows in a columnar structure following the van der Drift growth model.[28] This growth mode results in grains with varying crystallographic orientations across the film thickness, which poses challenges for surface processing and polishing due to the anisotropic mechanical properties at the grain boundaries.[35,36] Each diamond grain has a different crystallographic orientation, which affects its polishing resistance. Certain orientations, such as {111} planes, are significantly more difficult to polish than others, like {100} planes, leading to non-uniform surface roughness.[37] Grain boundaries contain non-diamond carbon phases, including amorphous and graphitic carbon, which have lower hardness and introduce defects. These inhomogeneous regions contribute to irregular polishing behavior, making it challenging to achieve a smooth surface. For successful direct bonding, the surface roughness of the PCD wafer must generally be reduced below 0.3 nm.[38–40] However, achieving this level of smoothness is an extremely difficult task due to the inherent grain boundary variations and differences in crystallographic orientations. Even if such ultra-smooth surfaces are obtained, the high cost associated with the polishing process presents a significant barrier to the industrial application of PCD wafers. As a result, alternative approaches are being explored, including advanced bonding technologies that can accommodate higher surface roughness,[24,25] enabling cost-effective GaN-on-diamond integration for high-power electronic applications. The direct bonding of PCD with high surface roughness remains a critical challenge, requiring the development of novel interface engineering strategies to enhance bonding strength, minimize thermal resistance, and ensure device reliability.

In this study, we use an advanced bonding technique to demonstrate the fabrication of GaN HEMTs on a 2-inch PCD wafer by transferring AlGaN/GaN/3C-SiC layers onto a high-surface-roughness PCD wafer. Following the layer transfer, GaN HEMTs were fabricated using a conventional process, incorporating a high-temperature annealing step to achieve ohmic contacts. To further evaluate thermal stability, an annealing process at 1100 °C was conducted on the AlGaN/GaN/3C-SiC/PCD bonded wafers to assess their structural integrity and thermal reliability. The surface roughness of PCD and the surface morphology of grains were analyzed using atomic force microscopy (AFM). The crystal orientation and grain size distribution of the PCD wafer were investigated using electron backscatter diffraction (EBSD). The thermal conductivity of both SCD and PCD wafers was measured from their growth surfaces using the time-domain thermoreflectance (TDTR) technique.

The bonding interface between 3C-SiC and PCD was examined using transmission electron microscopy (TEM) and energy-dispersive X-ray spectroscopy (EDS) to evaluate structural integrity and elemental distribution. Finally, the electrical properties and surface temperature during the operation of GaN HEMTs on PCD were compared with those of GaN HEMTs fabricated on Si, SiC, and SCD wafers. This comparison provides insights into the advantages of thermal management and performance improvements enabled by GaN-on-diamond integration, particularly in high-power and high-frequency applications.

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2. Experimental Section

The HEMT heterostructure used in this study consists of multiple layers grown on a 6-inch Czochralski (Cz)-Si (111) wafer via metal-organic chemical vapor deposition (MOCVD). The layer stack includes a 23-nm-thick $Al_{n\cdot 26}Ga_{n\cdot 74}N$ barrier layer, a 2-µm-thick GaN layer, a 1-µm-thick AlGaN/AlN buffer layer, and a 1-µm-thick 3C-SiC layer. The AlGaN/GaN/3C-SiC heterostructure was grown on the Si substrate by AIR Water Inc. using its proprietary high-vacuum epitaxial growth system.[41] Initially, the heterostructure was temporarily bonded to a carrier wafer, after which the Si wafer was removed using fluoric acid etching. The exposed 3C-SiC surface undergoes chemical mechanical polishing to achieve a flat surface, followed by cleaning with acetone and isopropyl alcohol and drying under an N_2 flow. After surface preparation, the 3C-SiC layer was bonded to the growth surface of PCD wafers of various sizes, including 10 × 10 mm, 1-inch, and 2-inch wafers, all with a thickness of 400 µm.

The PCD wafers were synthesized on Si substrates using microwave plasma chemical vapor deposition (MPCVD). Before diamond growth, the Si substrates were mechanically pre-treated by grinding with fine sandpaper and diamond powder (50–100 nm in diameter) to introduce sufficient nucleation sites for diamond growth. Following this, the substrates underwent an ultrasonic cleaning process using acetone, ethanol, and deionized water for 5 min each, effectively removing surface contaminants. The cleaned substrates were then dried with nitrogen before placement in the MPCVD chamber. The substrates were positioned on a tray inside the chamber. Once the chamber pressure reached below 10^{-5} Torr, hydrogen gas flow was initiated. Using 2000 W microwave power, the resulting hydrogen plasma fully cleaned the substrate and chamber for 15 min. A methane-hydrogen gas mixture was introduced, maintaining a methane volume fraction of 2–5%. The working gas pressure and microwave power were increased to 200 Torr and 5.6 KW, respectively. To optimize substrate temperature (700–1100 °C), the Z-axis height in the MPCVD system was adjusted to control the distance between the substrate and plasma. The gas flow rates were set to 10–25 sccm for methane and 400–490 sccm for hydrogen. The total deposition time for all samples was maintained at 400 h, ensuring uniform diamond layer formation. After completing PCD growth, the PCD-on-Si wafer was removed from the chamber and subsequently separated from the Si substrate. Before bonding, the growth surfaces of PCD wafers were mechanically polished and cleaned with a sulfuric acid-hydrogen peroxide mixture to ensure a pristine surface.

The bonding process was carried out at room temperature using the surface-activated bonding (SAB) method. During bonding, the growth surfaces of the diamond and 3C-SiC were simultaneously irradiated by an Ar fast atom beam (FAB) in a high vacuum pressure of 1.0×10^{-7} Pa. The Ar FAB irradiation voltage and current are 1.6 kV and 160 mA, respectively. Immediately after the irradiation processes, the diamond and 3C-SiC were brought into contact under a load of 1 GPa for 180 s at room temperature. Following bonding, the carrier wafer was removed, successfully transferring the AlGaN/GaN/3C-SiC layers onto the PCD wafer, facilitating further device fabrication and characterization. This transfer process has steadily progressed over the years. In 2020, the AlGaN/GaN/3C-SiC layers were first transferred onto the 10 × 10 mm PCD wafer. By 2022, the process was successfully scaled to the 1-inch PCD wafer. Finally, in 2025, the AlGaN/GaN/3C-SiC layers were successfully transferred onto the 2-inch CD wafer, representing a significant milestone in advanced large-area GaN-on-diamond technology, as shown in **Figure 1**. The AlGaN/GaN/3C-SiC layers were also successfully transferred onto a 4-inch 4H-SiC wafer with a thickness of 500 µm and a 1-inch SCD wafer with a thickness of 300 µm, using the same bonding method.

Following the transfer, the AlGaN/GaN/3C-SiC/PCD wafers undergo GaN HEMT fabrication using a conventional process. The fabrication steps include mesa isolation via Ar ion milling and dry etching, followed by the formation of source and drain ohmic contacts through Ti/Al/Ti/Au multilayer evaporation and rapid thermal annealing at 800 °C for 60 s in an N_2 ambient. Finally, gate Schottky contacts were formed by evaporating Ni/Au layers, completing the device fabrication. The same fabrication process was applied to GaN HEMTs on the 4H-SiC and 1-inch SCD wafers.[30] EBSD measurements confirmed the single-crystal nature of the SCD, as shown in Figures S1 and S2 (Supporting Information). In SCD, ≈8 ppm of nitrogen was incorporated spontaneously during diamond growth. In contrast, GaN HEMTs on Si wafers were manufactured using a standard fabrication approach.[31] The transfer process of the AlGaN/GaN/3C-SiC layers and the fabrication steps of GaN HEMTs on a 2-inch PCD wafer based on the bonding-first concept are illustrated in **Figure 2**.

The surface roughness of PCD was analyzed using AFM. The AFM measurements were conducted with a Shimadzu SPM-9600, providing high-resolution surface characterization to assess the topographical features and roughness of the PCD wafer. The crystal orientation and grain size distribution of the PCD were analyzed using EBSD. EBSD measurements were performed with a JEOL JSM-IT800 field emission scanning electron microscope (FE-SEM), operated in secondary electron detection (SED) mode at an acceleration voltage of 15 kV, standard (STD) observation mode, and large irradiation current mode to ensure high-resolution crystallographic characterization. The sample was positioned at a 70° tilt angle during EBSD analysis. The scanning parameters were as follows: Growth surface: Scanning area of 400.00 × 915.00 μm^2 and a step size of 2.50 µm. Nucleation surface: Scanning area of 22.80 × 68.85 μm^2 and a step size of 0.15 µm. These measurement conditions allowed for a detailed characterization of the crystallographic structure and grain distribution across different regions of the PCD sample.

The thermal conductivity of PCD and SCD wafers was measured using the TDTR technique. Before TDTR measurements, a ≈90 nm thick aluminum (Al) layer was deposited on the growth surfaces of the PCD and SCD as a transducer. TDTR, an ultrafast laser-based pump-probe technique, enables the assessment of the thermal properties of PCD and SCD.[42] The pump beam periodically heats the sample surface, while the probe beam detects temperature variations via thermoreflectance. The resulting signal, captured by a photodetector and processed through a lock-in amplifier, was fitted to an analytical heat transfer model to extract the thermal properties of the sample structure.[43]

The structure and chemical composition of the 3C-SiC/PCD bonding interface, both before and after annealing at 1100 °C for 10 min, were analyzed using TEM with a JEOL JEM-2200FS

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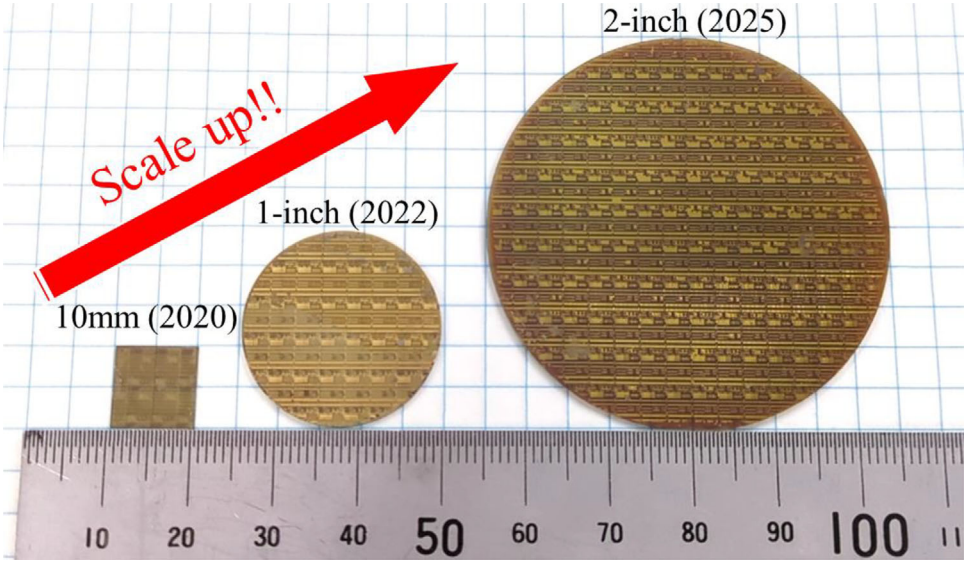


Figure 1. GaN HEMTs were fabricated on 10 mm × 10 mm PCD substrate, 1-inch PCD, and 2-inch PCD wafers.

analytical microscope at an acceleration voltage of 200 kV. At room temperature, TEM samples were prepared using a focused ion beam (FIB) technique with a Helios NanoLab 600i DualBeam system (Thermo Fisher Scientific). A Pt protective layer was deposited before milling to prevent damage during the FIB preparation. The FIB milling process was performed at an acceleration voltage of 30 kV, followed by final etching at 2 kV to ensure high-quality sample preparation with precise spatial resolution. Scanning transmission electron microscopy (STEM) analysis was conducted at 200 kV, providing detailed structural and compositional information on the 3C-SiC/PCD bonding interface. Energy-dispersive X-ray spectroscopy (EDS) was utilized to investigate the chemical composition, enabling elemental mapping and distribution analysis within the interface region.

The surface temperature of GaN HEMTs fabricated on Si, SiC, SCD, and PCD wafers was measured using the microphotoluminescence (μ-PL) technique. A 325 nm He-Cd laser with a power of 0.1 mW served as the excitation light source. The laser beam was focused on a 1 μm diameter spot using a 45× ultraviolet objective, and PL spectra were captured using a Si charge-coupled device (CCD) camera. To calibrate the PL spectra with ambient temperature, peak energies of the PL spectra were measured for GaN HEMTs under no bias at various temperatures. These data points were fitted using the Varshini equation,[44,45] a commonly

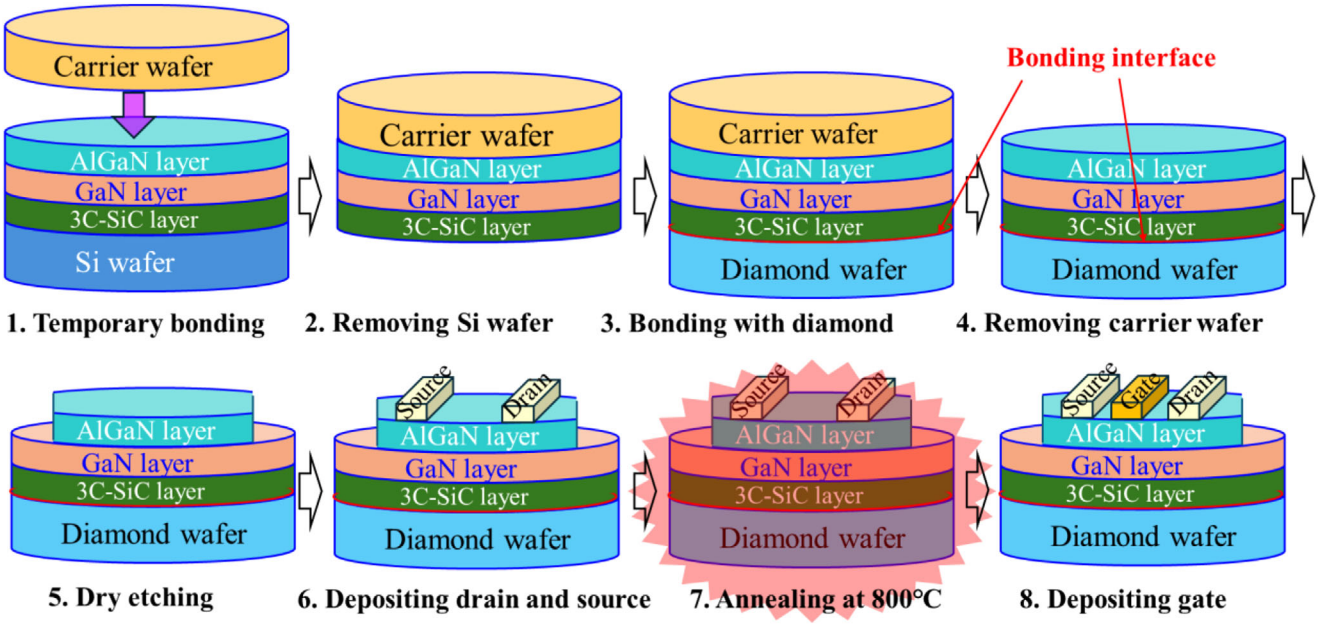


Figure 2. The transfer process of the AlGaN/GaN/3C-SiC layers and the fabrication step of GaN HEMTs on a 2-inch PCD wafer.

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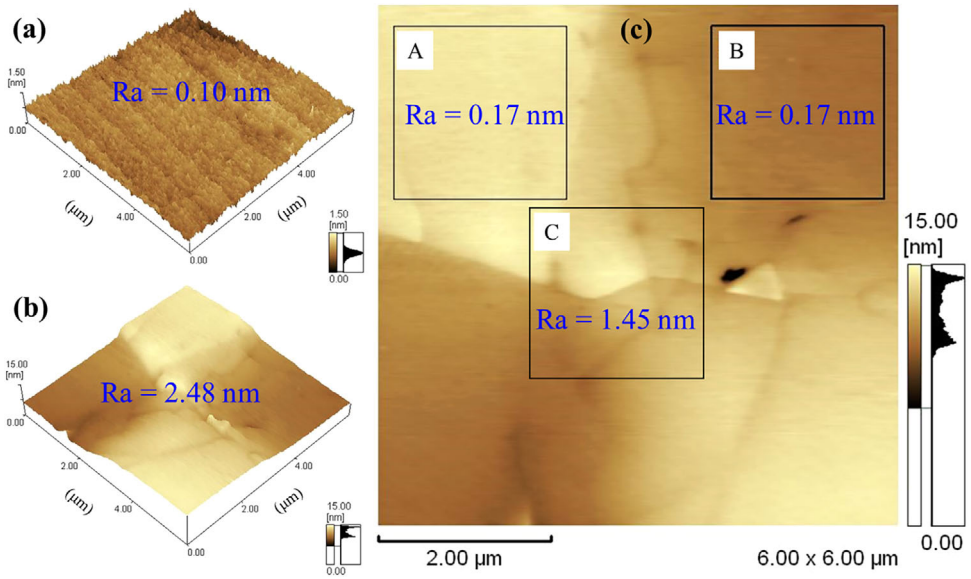


Figure 3. AFM 3D image of the PCD surface without (a) and with (b) grain boundaries, along with a 2D image highlighting the grain boundary region of the PCD surface (c).

used mathematical model for temperature calibration. The measurement accuracy had an associated error of ±5 °C. GaN HEMTs on Si, SiC, SCD, and PCD wafers were mounted on a copper heat sink for effective heat dissipation to ensure stable ambient temperature conditions. A water-cooling system actively cooled the heat sink, maintaining a fixed ambiance of 300 K throughout the measurements. The GaN HEMTs were biased at the same source-drain voltage while temperature data were recorded. A slight increase in current was observed due to the He-Cd laser, as shown in Figure S3 (Supporting Information). However, current-voltage curves recorded before and after the measurements confirmed that no shifts in device behavior occurred, indicating the laser had no long-term electrical impact on the devices. To assess potential laser-induced heating, the effect of different laser powers on the surface temperature of GaN HEMTs was systematically examined. No temperature rise attributable to the laser was detected, confirming that the experimental setup provided a stable and thermally controlled environment. This level of precise thermal regulation ensured the accuracy and reproducibility of temperature measurements across different GaN HEMT wafers.

3. Results and Discussion

Figure 3a–c presents AFM 3D and 2D images of the PCD surface, illustrating the impact of grain boundaries on surface roughness. Figure 3a depicts a highly polished PCD surface with minimal surface variations, exhibiting an average roughness (Ra) of 0.1 nm. The absence of grain boundaries contributes to this ultra-smooth morphology, indicating a nearly atomically flat surface. In contrast, Figure 3b shows a rougher PCD surface where grain boundaries are present. The Ra value increases significantly to 2.48 nm, highlighting the influence of grain boundaries on surface topography. These boundaries introduce surface irregularities due to differential polishing rates, localized stress, and material heterogeneity. Figure 3c focuses on the surface roughness variations around grain boundaries, visually emphasizing their impact on surface uniformity. The Ra value in these regions is notably higher than in areas without grain boundaries, with roughness at least an order of magnitude greater. This difference suggests that grain boundaries significantly contribute to increased surface irregularities, affecting the polishing efficiency and overall smoothness of PCD. The morphological contrast between Figure 3a,b demonstrates the challenge of achieving a uniformly smooth PCD surface in the presence of grain boundaries.

The image quality (IQ) map, the inverse pole figure (IPF) map, the color-coded map, the [111] direction pole figure corresponding to the IPF map, and the grain size distribution of the growth surface of PCD analyzed by transmission EBSD are shown in **Figure** 4a–e, respectively. The IQ map reveals discrete grains with homogeneous color, while the surrounding grain boundaries appear darker due to their lower IQ values. The IPF map provides further crystallographic information, displaying the orientation of individual grains relative to a reference direction. The color-coded map enhances this visualization, showing distinct grains with varying orientations. The [111] direction pole figure reveals no significant preferred crystallographic orientation, suggesting that the grain growth direction is uniformly distributed across the PCD surface. This isotropic crystal orientation can be advantageous for applications requiring uniform thermal and mechanical properties across the wafer, as it minimizes anisotropic variations that could affect performance. Grain size distribution analysis of the growth surface of the PCD sample shows a broad grain size range, from 7.5 to 141 µm, with the smallest grain measuring 7.5 µm and the largest reaching 141 µm. The average grain diameter is 55.9 µm, indicating a predominantly coarse-grained structure. Notably, 95% of the grains are larger than 10 µm, further confirming the coarse-grained nature of the growth surface. The larger grains are expected to enhance thermal conductivity, as reduced grain boundary density minimizes phonon scattering, to facilitate efficient heat dissipation. Based on the measured

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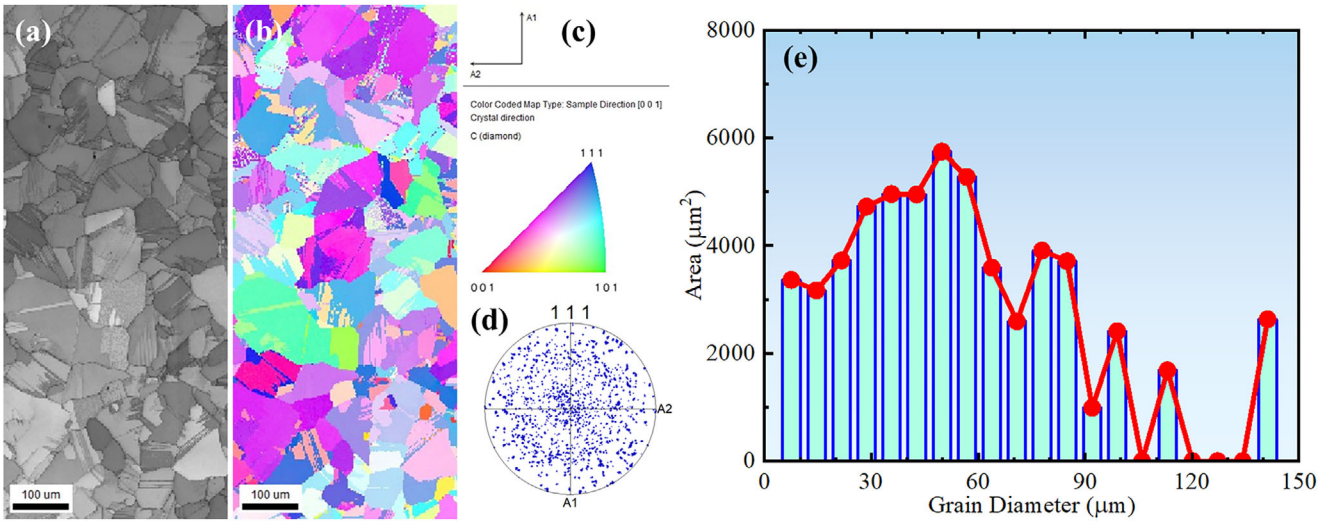


Figure 4. The IQ map (a), the IPF map (b), the color-coded map (c), the [111] direction pole figure corresponding to the IPF map (d), and the grain size distribution of the growth surface of PCD analyzed by transmission EBSD (e).

grain size distribution and the known correlation between grain size and thermal conductivity in PCD,[46–49] the thermal conductivity of the growth surface of PCD is predicted to exceed 2000 W m^{-1}·K. These findings highlight the diverse grain morphology within the PCD material, which influences its thermal properties. The uniform grain orientation and relatively large grain sizes suggest this PCD wafer could be highly suitable for high-power electronic applications, such as heat-spreading substrates for GaN HEMTs. The high thermal conductivity and efficient heat dissipation characteristics make it an ideal candidate for advanced thermal management solutions in next-generation power electronics.

Figure 5a–e presents the IQ map, IPF map, color-coded map, the [111] direction pole figure corresponding to the IPF map, and the grain size distribution of the nucleation surface of PCD analyzed using transmission EBSD. The color-coded map reveals a more refined grain structure than the growth surface. The [111] direction pole figure indicates no significant preferred crystallographic orientation, suggesting that the grain growth is more uniformly distributed across the PCD nucleation surface than the growth surface. The grain sizes on the nucleation surface exhibit a broad distribution, ranging from 0.38 to 5.85 μm. Specifically, the smallest grain on the nucleation surface measures 0.38 μm, while the largest grain reaches 5.85 μm. The average grain diameter is 2.2 μm, 1/25th of the average grain size on the growth surface, highlighting a significant contrast in grain structure. 98% of the grains are smaller than 5 μm, indicating a predominantly fine-grained structure. The presence of smaller grains

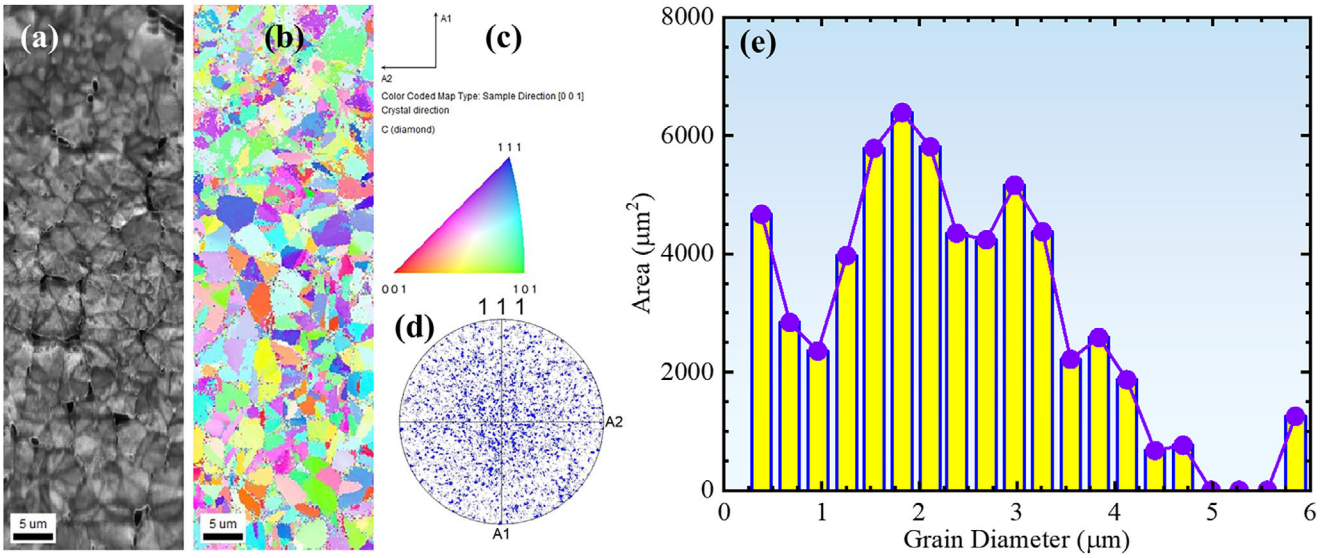


Figure 5. The IQ map (a), the IPF map (b), the color-coded map (c), the [111] direction pole figure corresponding to the IPF map (d), and the grain size distribution of the nucleation surface of PCD analyzed by transmission EBSD (e).

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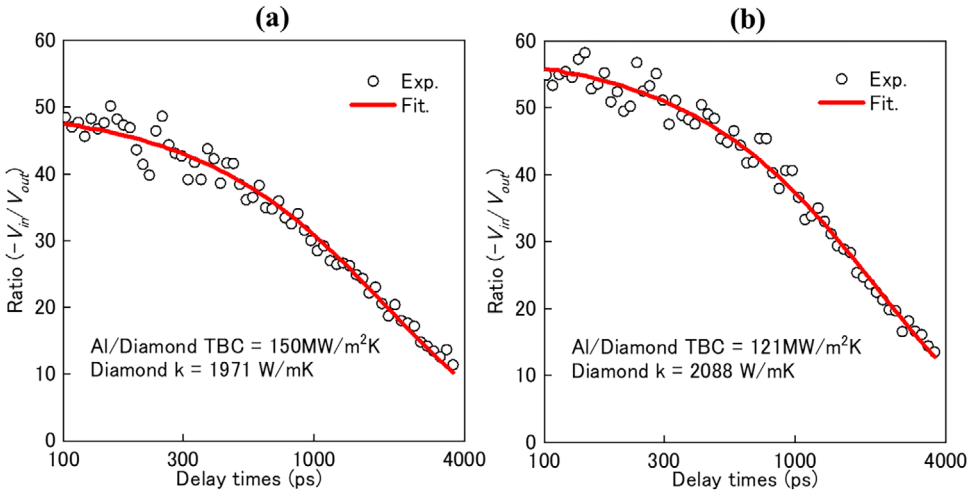


Figure 6. The data fitting of TDTR experiment data on SCD (a) and PCD (b) wafers.

leads to a reduction in thermal conductivity, as phonon scattering at grain boundaries increases, thereby impeding heat dissipation. Based on the measured grain size distribution and the established relationship between grain size and thermal conductivity in PCD,[46–49] the thermal conductivity of the nucleation surface is predicted to be smaller than 1000 W m^{-1}·K. However, this value remains lower than the growth surface, where large grains facilitate more efficient heat transport. These findings underscore the critical importance of grain size optimization in enhancing the thermal properties of PCD-based materials. Controlling grain growth and minimizing phonon scattering effects to improve the overall heat dissipation performance, making PCD a promising candidate for high-power electronic applications that demand efficient thermal management.

The TDTR fitting data for the experimental ratio signal and analytical heat transfer solution of the Al-SCD and Al-PCD structures are shown in **Figure** 6a,b. The extracted thermal properties reveal that SCD has a thermal conductivity of 1971 W m^{-1}·K, with a thermal boundary conductance (TBC) of 150 MW m^{-2}·K at the Al-SCD interface. SCD exhibits isotropic thermal conductivity,[50] meaning its heat transport properties remain uniform in all directions. The SCD used in this work contains nitrogen impurities, which reduce its thermal conductivity compared to pure diamond single crystals. The nitrogen-related defects introduce phonon scattering, thereby limiting heat transport efficiency. In contrast, PCD exhibits a thermal conductivity of 2088 W m^{-1}·K but a lower TBC of 121 MW m^{-2}·K at the Al-PCD interface. Unlike SCD, PCD shows anisotropic thermal conductivity due to grain size and orientation variations.[49] Additionally, the measured thermal conductivity primarily represents the thermal properties near the growth surface rather than those of the entire bulk material. These results highlight the differences in heat dissipation efficiency between SCD and PCD. In our TDTR thermal conductivity measurements of both SCD and PCD wafers, we estimate the experimental uncertainty to be ≈±10%, which is consistent with previously reported values for high-thermal-conductivity materials measured using the TDTR technique.[51] It should be noted that the SCD used in this study does not represent the highest-quality SCD reported in the literature, where thermal conductivity can exceed 3000 W m^{-1}·K. Nevertheless, the comparison provides a useful reference point for evaluating the effectiveness of PCD in GaN-on-diamond applications.

The higher effective thermal conductivity of PCD compared to SCD can be attributed to its large grain diameter and lower impurity concentrations. While SCD may contain impurities or vacancies that scatter phonons and reduce thermal conductivity,[47,50,52] PCD benefits from grain boundary engineering. When the grain size is sufficiently large, phonon scattering at grain boundaries is minimized,[46,48] allowing for more efficient heat transport. In PCD wafers, ≈95% of the grains exceed 10 μm in size, which aligns with the observed reduction in phonon scattering and contributes to the enhanced thermal conductivity of PCD. These characteristics make PCD a highly effective material for high-performance thermal management applications, particularly in GaN-based high-power electronics, where efficient heat dissipation is critical for device reliability and longevity.

Figure 7a–d presents a cross-sectional low-magnification TEM image and high-resolution TEM (HRTEM) image of the 3C-SiC/PCD interface before annealing, EDS elemental mappings of the interface, and X-ray intensity profiles for silicon, carbon, oxygen, and argon atoms across the interface. TEM analysis reveals the formation of an intermediate layer at the interface with a measured thickness of ≈7 nm. However, the intermediate layer exhibits nonuniform thickness, ranging from 5 nm in thinner regions to 8 nm in thicker areas. The variation in thickness suggests surface roughness differences on the bonding surface, which aligns with observed surface roughness variations in PCD. Notably, the PCD wafer, with a surface roughness of 2.48 nm, was successfully bonded to 3C-SiC, demonstrating the feasibility of achieving strong interfacial bonding despite surface roughness variations. This highlights the importance of precise surface preparation in achieving high-quality bonding between PCD and 3C-SiC, which is critical for enhancing thermal and mechanical performance in high-power electronic applications.

To further investigate the structure of this layer, fast Fourier transform (FFT) analysis was applied, confirming its amorphous nature and indicating a lack of long-range atomic order. The formation of this amorphous layer is attributed to argon fast atom

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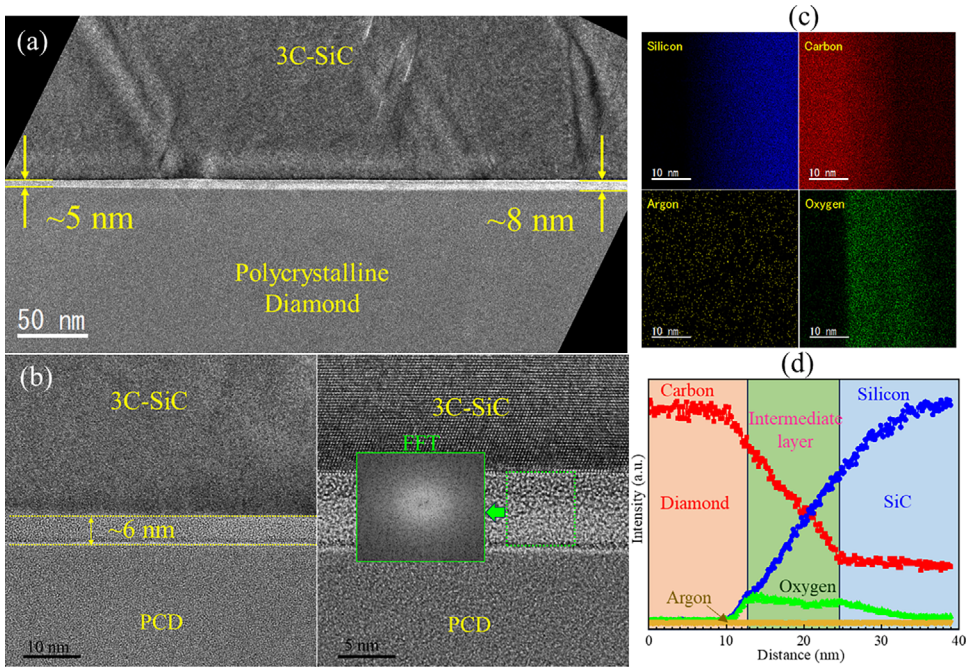


Figure 7. A cross-sectional low-magnification TEM (a) and HRTEM (b) images of the 3C-SiC/PCD interface before annealing, EDS elemental mappings of the interface (c), and X-ray intensity profiles for silicon (blue), carbon (red), oxygen (green), and argon (dark yellow) atoms across the interface (d). The FFT image is overlayed in Figure 6b.

beam (FAB) irradiation during the bonding process, which likely induced surface modifications leading to interface amorphization. Similar amorphous intermediate layers have been reported in the interface systems fabricated using the SAB method, including Si/SiC,[53] Si/Si,[54] GaAs/SiC,[55] and Si/Ga_2O_3.[56] EDS elemental mapping confirms the presence of silicon, carbon, oxygen, and argon at the interface. Corresponding X-ray intensity profiles provide quantitative insights into their distribution across the bonded interface. The detected oxygen is likely attributed to residual contamination from the vacuum system, whereas the presence of argon is associated with FAB irradiation. However, the argon concentration was below the resolution threshold of the EDS detector, making precise quantification challenging.

Figure 8a–d shows the cross-sectional low-magnification (TEM) image and HREM image of the 3C-SiC/PCD interface after annealing at 1100 °C, EDS elemental mappings of the interface, and X-ray intensity profiles for silicon, carbon, oxygen, and argon atoms across the interface. TEM analysis reveals the formation of an intermediate layer at the interface with a uniform thickness of ≈13 nm. Compared to the interface before annealing, this intermediate layer exhibits greater uniformity and increased thickness, suggesting structural evolution induced by high-temperature annealing. The growth in thickness is likely attributed to atomic diffusion and phase transformation occurring at the interface. Grains with long-range atomic order were observed within this intermediate layer, corresponding to the 3C-SiC {101} planes. The presence of these ordered structures suggests partial crystallization of the previously amorphous interface, indicating an annealing-induced transition toward a more thermodynamically stable phase. A similar transformation has also been observed at the 3C-SiC/SCD and Si/SCD interfaces after annealing at high temperatures,[30,56,57] further supporting the role of annealing in promoting structural evolution and crystallization at the bonded interface. This transformation results in the forming of a SiC polycrystalline layer at the interface.

EDS elemental mapping confirms the distribution of silicon, carbon, oxygen, and argon within the interface region. At the same time, the corresponding X-ray intensity profiles provide quantitative insights into elemental diffusion across the bonded interface. Argon atoms were not overlayed in Figure 8d due to their intensity being lower than the resolution threshold of the EDS detector. Additionally, oxygen atoms were concentrated at the interface adjacent to the SiC layer, suggesting localized oxidation or residual contamination. This localized oxidation may contribute to an increase in TBC at the interface. However, this effect can be minimized by improving vacuum conditions during the bonding process, thereby reducing oxygen contamination and enhancing interfacial thermal performance. The observed diffusion behavior indicates that annealing promotes atomic rearrangement, leading to interfacial composition and microstructure. This transformation contributes to increased crystallinity and redistribution of chemical elements within the interface. These findings highlight the significant impact of high-temperature annealing on the interfacial properties of the bonded structure, enhancing crystallization while altering elemental distribution. Such changes may influence the thermal and electrical characteristics of the interface, which are critical for optimizing the performance and long-term reliability of bonded 3C-SiC/PCD structures in semiconductor applications.

The drain current (I_D)–drain bias voltage (V_{DS})–temperature characteristics of GaN HEMTs fabricated on Si, SCD, and PCD

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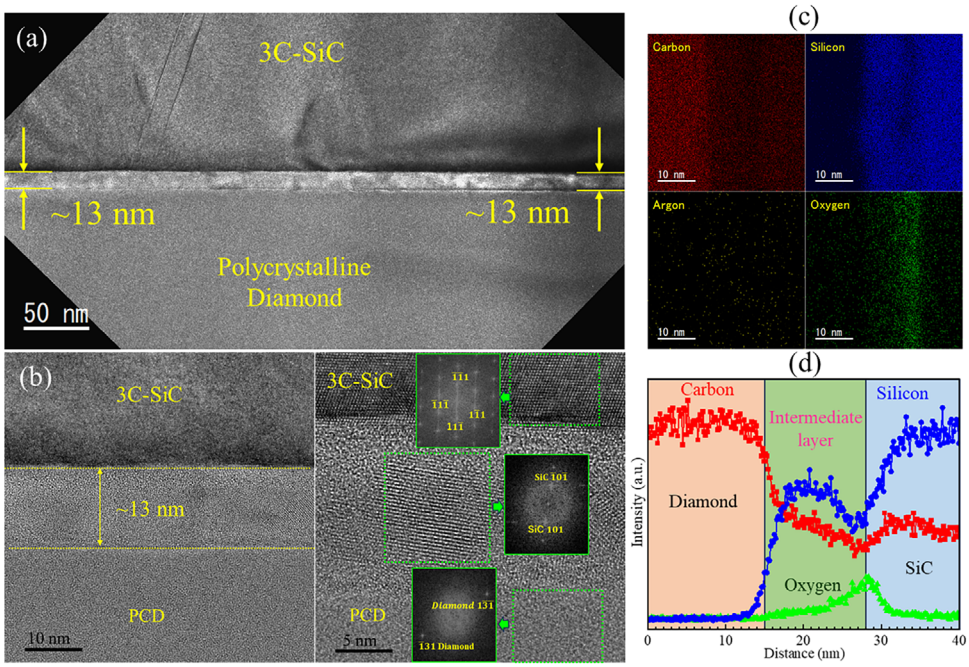


Figure 8. A cross-sectional low-magnification TEM (a) and HRTEM (b) images of the 3C-SiC/PCD interface after annealing at 1100 °C, EDS elemental mappings of the interface (c), and X-ray intensity profiles for silicon (blue), carbon (red), oxygen (green), and argon (dark yellow) atoms across the interface (d). The FFT image is overlayed in Figure 7b.

wafers were investigated. The GaN HEMTs have a gate length (L_G) of 5 µm, gate width (W_G) of 300 µm, source–gate spacing (L_{SG}) of 10 µm, and gate-drain spacing (L_{GD}) of 10 µm. **Figure** 9a,b illustrates the schematic diagram of the gate edge temperature measurement in GaN HEMTs using the µ-PL technique and the relationship between the temperature rise at the gate edge under different power dissipation (P_{diss}) values (2, 4, 6, 8, 10 W mm^{-1}) for GaN HEMTs on Si, SCD, and PCD wafers. Before power application, the temperatures in the gap between the gate and drain of GaN HEMTs on Si, SCD, and PCD wafers were ≈27 °C. The most significant temperature rise occurs at the gate edge within this gap, where a high electric field is applied.[58,59]

Upon applying a P_{diss} of 8 W mm^{-1}, the temperature increases at the gate edge varied significantly depending on the substrate material: GaN HEMTs on Si increased to 258.8 °C (ΔT = 231.8 °C), GaN HEMTs on SCD increased to 81 °C (ΔT = 54.0 °C), GaN HEMTs on PCD increased to 109.8 °C (ΔT = 73.8 °C). These results highlight diamond-based substrates' superior thermal dissipation properties, where SCD and PCD wafers exhibited significantly lower temperature rises than Si, demonstrating their effectiveness in heat management. The temperature rise at the gate edge of GaN HEMTs is proportional to P_{diss}, allowing the extraction of thermal resistance (R_{TH}) based on the slope of P_{diss} versus temperature rise (ΔT). The extracted R_{TH} value for

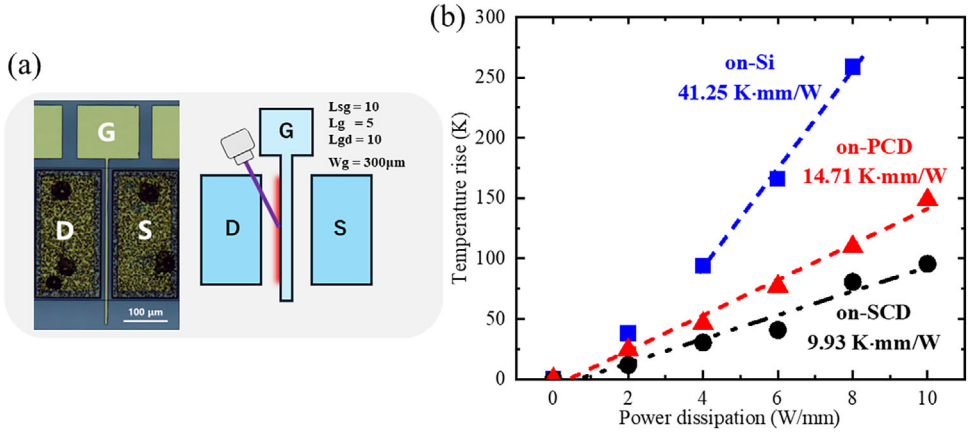


Figure 9. Schematic diagram of the gate edge temperature measurement in GaN HEMTs (a) and the relationship between the temperature rise at the gate edge and P_{diss} for GaN HEMTs on Si, SCD, and PCD wafers (b).

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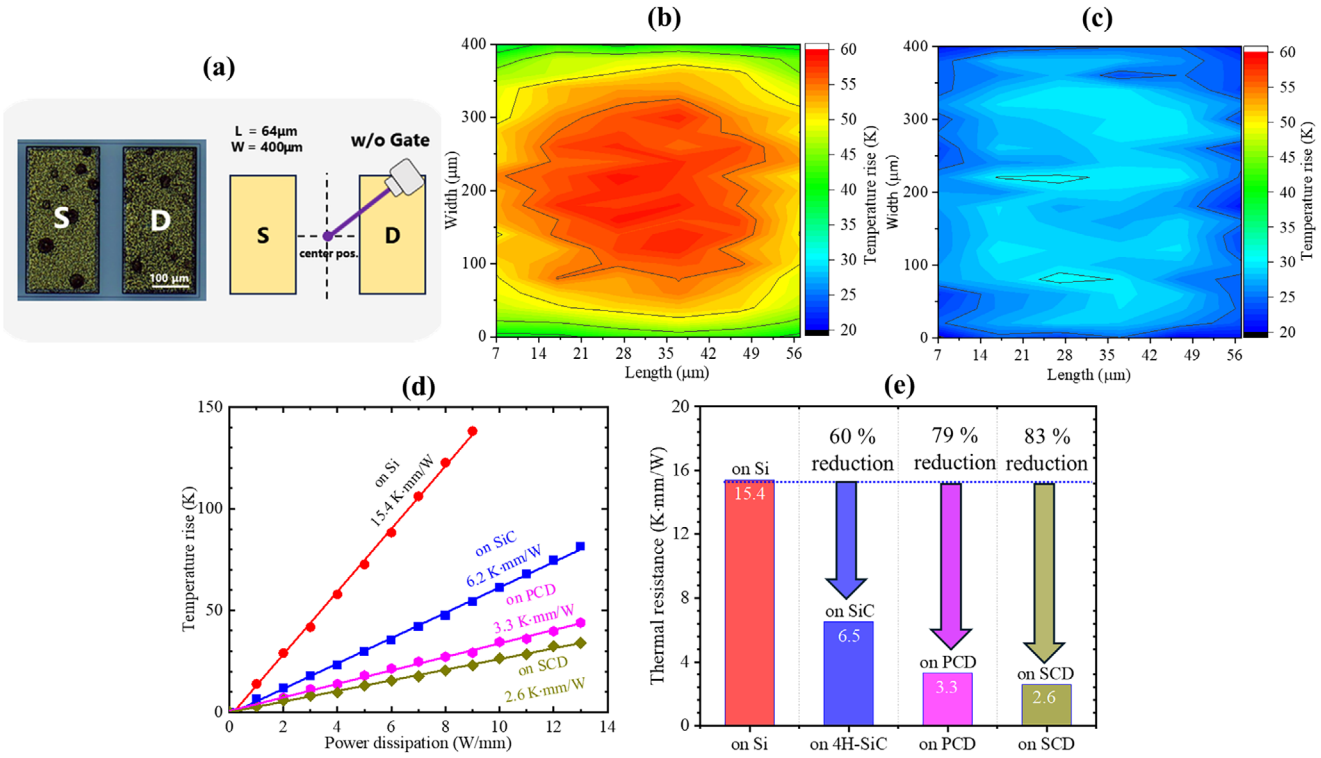


Figure 10. Schematic diagram of the temperature measurement setup in the region between the source and drain of GaN HEMTs without a gate electrode (a), temperature distribution mapping in the space between the source and drain of GaN HEMTs without a gate electrode on Si (b) and PCD (c) for P_{diss} = 4 W mm^{-1}, the relationship between the temperature rise at the center position between the source and grain for GaN HEMT without a gate electrode on Si, SiC, SCD, and PCD wafers with applied different P_{diss} (d), and the extracted R_{TH} of GaN HEMT without a gate electrode on different wafers (e).

different substrates is 41.3 K·mm W^{-1} of GaN HEMTs on Si, 9.9 K·mm W^{-1} of GaN HEMTs on SCD, and 14.7 K·mm W^{-1} of GaN HEMTs on PCD. The significantly lower R_{TH} values of GaN HEMTs on SCD and PCD compared to Si demonstrate their strong potential for high-power applications where efficient thermal management is critical.

Heat dissipation in GaN HEMTs is strongly influenced by device geometry, as it dictates how heat is generated and distributed across the structure.[60,61] The highest temperature rise typically occurs at the gate edge in the gate-drain region, where the electric field is strongest—smaller gate-to-drain spacing results in a higher power density, leading to more localized heating. Increasing gate-to-drain spacing can improve heat spreading and reduce localized thermal stress, but may impact RF performance. Shorter L_G leads to higher power densities due to smaller channel areas, increasing self-heating effects. Wider W_G allows for better heat dissipation, as heat can spread more effectively across the device. To further investigate the heat dissipation performance of GaN-on-diamond structures, the drain-source bias current-voltage (I-V) characteristics of GaN HEMTs without a gate electrode, fabricated on Si, SiC, PCD, and SCD wafers, were analyzed. This approach ensures that the spacing between the source and drain is the primary heating region, allowing for a clear assessment of heat dissipation efficiency across different substrates. The drain-source spacing was 64 μm in length and 400 μm in width, providing a standardized basis for comparison.

Figure 10a–c shows the schematic diagram of the temperature measurement setup in the region between the source and drain of GaN HEMTs without a gate electrode, the temperature Distribution in this region for GaN HEMTs on Si and PCD wafers at Pdiss = 4 W mm^{-1}, the relationship between the temperature rise at the center position between the source and drain for GaN HEMT on Si, SiC, PCD, and SCD wafers under various P_{diss}, and the extracted R_{TH} of GaN HEMT on different wafers. The thermal mapping of GaN HEMTs reveals significant differences in heat dissipation efficiency between Si and PCD wafers. GaN HEMTs on Si wafers exhibit higher temperatures, with heat concentrated at the center of the region between the source and drain, indicating less efficient heat dissipation. GaN HEMTs on PCD wafers show a lower and more uniform temperature distribution, demonstrating the superior thermal conductivity of PCD and its ability to effectively spread heat across the device.

The temperature at the center position between the source and grain of GaN HEMT increases linearly with P_{diss}, indicating a direct correlation between thermal load and substrate heat dissipation efficiency. The extracted R_{TH} values for GaN HEMTs on Si, SiC, PCD, and SCD are 15.4, 6.2, 3.3, and 2.6 K·mm W^{-1}, respectively. Compared to GaN HEMTs on Si, the R_{TH} values decreased by 60%, 79%, and 83% for SiC, PCD, and SCD wafers, respectively. These reductions highlight the exceptional thermal dissipation capabilities of diamond-based substrates, with SCD and PCD exhibiting the lowest thermal resistance. The significant decrease in R_{TH} for SCD and PCD demonstrates their high potential for effective thermal management in high-power GaN

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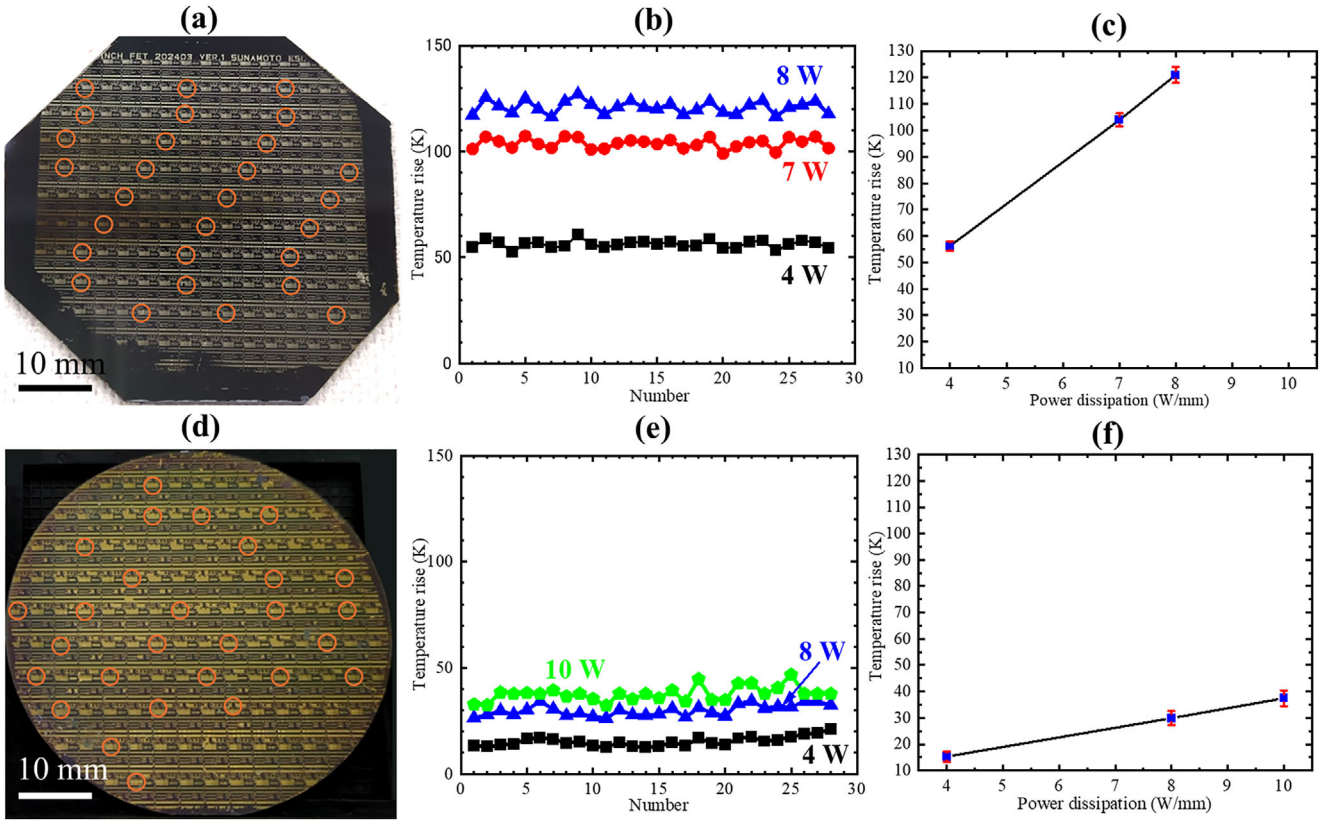


Figure 11. Measured positions (indicated by circles) of 28 GaN HEMTs without a gate electrode on 2-inch Si (a) and PCD (d) wafers, temperature rise uniformity of 28 GaN HEMTs on Si (b) and PCD (e) under various P_{diss}, and mean and standard deviation of the temperature rise for 28 GaN HEMTs without a gate electrode on Si (c) and PCD (f) under various P_{diss}.

HEMTs, helping to mitigate device self-heating and enhance overall performance. Their high thermal conductivity and superior heat-spreading properties make them ideal candidates for advanced thermal management solutions, particularly in high-power electronic applications, where minimizing thermal resistance is crucial to ensuring optimal device operation, reliability, and longevity.

In Figure 9, the HEMT devices were measured under gate-controlled conditions with V_{GS} fixed at −1 V and V_{DS} swept from 0 V to several tens of volts to achieve the desired power density. This setup reflects practical device operation, where localized heat is generated near the drain-side edge of the gate, rather than in an open-channel state. In contrast, the TLM structures in Figure 10 were measured under conditions resembling open-channel conduction, resulting in broader, planar heat dissipation. While the heat source in HEMTs can be approximated as a narrow line, in TLMs it is more accurately represented as a planar source. We examined both configurations to evaluate how the heat source geometry affects the thermal spreading behavior of the diamond layer. As device integration and miniaturization progress, densely packed line-shaped heat sources will effectively act as planar heat sources, making planar heat dissipation highly relevant for future device architectures. Additionally, TLM structures allow evaluation of the intrinsic thermal resistance of the diamond heat spreader without the influence of device bias conditions, which can significantly affect the temperature rise in HEMTs even at the same power density. These distinctions help clarify the thermal performance advantages of the diamond layer under different operational regimes.

Figure 11a–f illustrates the measured positions of 28 GaN HEMTs without a gate electrode on 2-inch Si and PCD wafers, the temperature rise uniformity of 28 GaN HEMTs Si and PCD under various P_{diss} values, and the mean and standard deviation of the temperature increase for 28 GaN HEMTs on Si and PCD wafers under different P_{diss} values. The temperature rise distribution was analyzed across 28 GaN HEMTs on 2-inch Si and PCD wafers under different P_{diss} values. Si wafers display a high-temperature rise with increasing P_{diss}, indicating inefficient heat dissipation and increased thermal hotspots. PCD wafers exhibit a lower temperature rise and a more uniform thermal distribution, indicating enhanced heat dissipation and consistent thermal performance across the wafer. Additionally, the standard deviation values of the temperature rise for 28 GaN HEMTs on PCD are comparably small to those observed for Si wafers under different P_{diss} values. This suggests that PCD wafers provide stable and uniform thermal characteristics, reinforcing their suitability for high-power GaN HEMT applications. These results confirm that PCD significantly enhances heat dissipation in GaN HEMTs, effectively reducing localized heating and improving thermal uniformity across the 2-inch wafer. The lower temperature rise and more uniform heat distribution on PCD wafers establish them as a highly promising substrate material for high-power GaN

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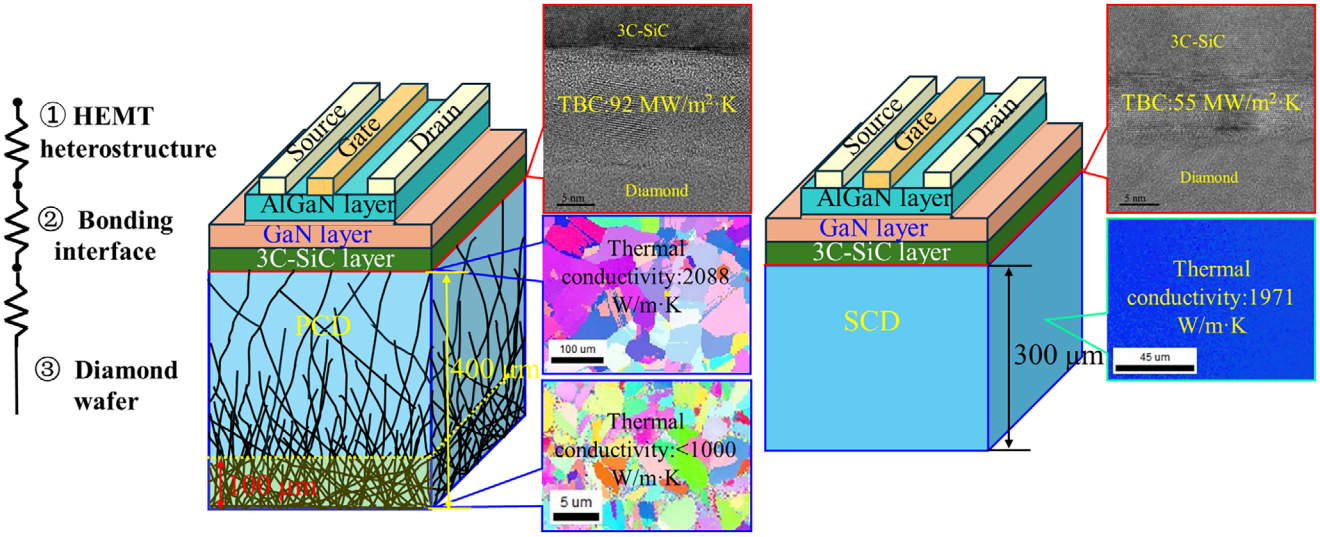


Figure 12. Thermal resistance components of GaN HEMTs on PCD and SCD wafers. The average grain diameter on the growth surface of the PCD wafer is significantly larger than that on the nucleation surface.

HEMT applications, where efficient thermal management is critical for device performance and reliability.

Figure 12 illustrates the thermal resistance components of GaN HEMTs on PCD and SCD wafers. These components consist of HEMT heterostructure, bonding interface, and diamond wafer. Since the same HEMT heterostructure is used in this study, the key difference in thermal performance arises from the bonding interface and the diamond substrate. Notably, the TBC at the 3C-SiC/PCD interface was measured at 92 MW m^{-2}·K, higher than the 55 MW m^{-2}·K reported for the 3C-SiC/SCD interface.[62] However, this value is still lower than the TBC observed at the 4H-SiC/3C-SiC/diamond interface, where the 3C-SiC layer was as thin as 5 nm.[63,64] This comparison suggests that reducing the thickness of the 3C-SiC layer at the interface could further enhance the TBC at the 3C-SiC/PCD interface. Additionally, the thermal conductivity of the growth surface on PCD is higher than that of SCD, suggesting superior heat transport properties in PCD. Despite these advantages, the R_{TH} of GaN HEMTs on PCD is 48% higher than that of GaN HEMTs on SCD. Similarly, for GaN HEMTs without a gate electrode, the R_{TH} on PCD is 27% higher than that on SCD. This discrepancy can be attributed to the contrast in grain structure between the nucleation and growth surfaces of PCD. The average grain size on the nucleation surface is significantly smaller than on the growth surface, leading to increased phonon scattering at grain boundaries. This enhanced scattering effect contributes to higher thermal resistance, despite PCD's intrinsically high thermal conductivity.

The coarse-grained growth surface facilitates efficient heat transport, while the fine-grained nucleation surface introduces additional thermal resistance, acting as a thermal bottleneck that limits overall heat dissipation efficiency. This highlights the critical role of grain size distribution in determining the overall thermal performance of PCD-based substrates. Further grain size optimization is necessary for fully leveraging the thermal advantages of PCD for high-power GaN HEMT applications. One potential approach to improving the thermal performance of PCD substrates is to remove the fine-grained nucleation layer using mechanical polishing techniques. This process can effectively reduce phonon scattering, enhance heat transport, and ultimately lower the thermal resistance of PCD. By removing the small-grain layer (≈100 μm thick), a higher thermal conductivity PCD can be obtained, allowing for more efficient heat dissipation. Refining grain structure control and minimizing phonon scattering at grain boundaries are essential strategies for optimizing PCD's thermal properties. By eliminating the fine-grained layer, heat conduction pathways can be significantly improved, making PCD-based substrates highly suitable for high-power electronic applications that demand efficient thermal management and long-term reliability. This enhancement is particularly beneficial for GaN HEMTs and other high-power semiconductor devices, where superior heat dissipation is crucial for ensuring optimal performance, stability, and longevity.

4. Conclusion

GaN HEMTs on a 2-inch PCD wafer were successfully fabricated through the direct integration of a PCD wafer, with a surface roughness of 2.48 nm, to 3C-SiC at room temperature using an advanced bonding technique. Before annealing, a nonuniform amorphous layer with a thickness of ≈7 nm was observed at the interface. After annealing at 1100 °C, this amorphous layer underwent a phase transformation, forming a uniform polycrystalline SiC layer with a thickness of ≈13 nm. No cracks and separations were observed at the interface. The thermal conductivity of the growth surface of PCD is higher than that of SCD, and the TBC at the 3C-SiC/PCD interface is higher than that at the 3C-SiC/SCD interface. However, the thermal resistance (R_{TH}) of GaN HEMTs on PCD was 48% higher than that of GaN HEMTs on SCD. For GaN HEMTs without a gate electrode, R_{TH} on PCD was 27% higher than on SCD. This discrepancy was attributed to the smaller grain size on the nucleation surface of PCD, which results in lower thermal conductivity. Despite the higher thermal conductivity of the growth surface, the fine-grained nucleation surface of PCD introduces phonon scattering, increasing thermal

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resistance. GaN HEMTs on polycrystalline diamond (PCD) exhibited a lower temperature rise and a more uniform thermal distribution, demonstrating enhanced heat dissipation efficiency and consistent thermal performance across the wafer. The coarse-grained growth surface enhances thermal transport, while the fine-grained nucleation surface acts as a thermal bottleneck. To fully leverage PCD's thermal advantages, further grain size optimization and interface engineering are required. Mechanical polishing techniques can help remove the fine-grained nucleation layer, reducing phonon scattering and improving thermal transport efficiency. The successful direct bonding of PCD with 3C-SiC, achieved with a roughness value, marks a significant milestone in electronics thermal management technology. This breakthrough enhances the practical industrial application of PCD by improving heat dissipation efficiency, reducing thermal resistance, and enabling high-power device reliability. As a result, PCD-based solutions are now poised to play a key role in the future of high-performance electronics, ensuring higher power densities, longer device lifetimes, and improved operational stability.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

3C-SiC integration, direct bonding, GaN high-electron-mobility transistors (HEMTs), polycrystal diamond (PCD), thermal management

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[1] J. Y. Tsao, S. Chowdhury, M. A. Hollis, D. Jena, N. M. Johnson, K. A. Jones, R. J. Kaplar, S. Rajan, C. G. Van de Walle, E. Bellotti, C. L. Chua, R. Collazo, M. E. Coltrin, J. A. Cooper, K. R. Evans, S. Graham, T. A. Grotjohn, E. R. Heller, M. Higashiwaki, M. S. Islam, P. W. Juodawlkis, M. A. Khan, A. D. Koehler, J. H. Leach, U. K. Mishra, R. J. Nemanich, R. C. N. Pilawa-Podgurski, J. B. Shealy, Z. Sitar, M. J. Tadjer, et al., *Adv. Electron. Mater.* **2018**, *4*, 1600501.
[2] A. J. Green, J. Speck, G. Xing, P. Moens, F. Allerstam, K. Gumaelius, T. Neyer, A. Arias-Purdue, V. Mehrotra, A. Kuramata, K. Sasaki, S. Watanabe, K. Koshi, J. Blevins, O. Bierwagen, S. Krishnamoorthy, K. Leedy, A. R. Arehart, A. T. Neal, S. Mou, S. A. Ringel, A. Kumar, A. Sharma, K. Ghosh, U. Singisetti, W. Li, K. Chabak, K. Liddy, A. Islam, S. Rajan, et al., *APL Mater* **2022**, *10*, 029201.
[3] G. Pavlidis, A. M. Hilton, J. L. Brown, E. R. Heller, S. Graham, *J. Appl. Phys.* **2020**, *128*, 075705.
[4] F. Wilhelmi, Y. Komatsu, S. Yamaguchi, Y. Uchida, T. Kase, S. Kunori, A. Lindemann, *IEEE Trans. Power Electron.* **2023**, *38*, 2107.
[5] A. L. Moore, L. Shi, *Mater. Today* **2014**, *17*, 163.
[6] Z. He, Y. Yan, Z. Zhang, *Energy* **2021**, *216*, 119223.
[7] J. E. Graebner, S. Jin, G. W. Kammlott, J. A. Herb, C. F. Gardinier, *Nature* **1992**, *359*, 401.
[8] J. S. Kang, M. Li, H. Wu, H. Nguyen, T. Aoki, Y. Hu, *Nat. Electron.* **2021**, *4*, 416.
[9] M. M. Sabry, A. Sridhar, D. Atienza, Y. Temiz, Y. Leblebici, S. Szczukiewicz, N. Borhani, J. R. Thome, T. Brunschwiler, B. Michel, *Proc.-Des., Autom. Test Eur.* **2011**, 1466.
[10] M. S. Shamim, R. S. Narde, J. L. Gonzalez-Hernandez, A. Ganguly, J. Venkatarman, S. G. Kandlikar, *Sustain. Comput.: Inf. Syst.* **2019**, *21*, 165.
[11] Q. Wang, Y. Wu, S. Niu, X. Zhao, *Energies* **2022**, *5*, 3249.
[12] W. Dong, X. Zhang, B. Liu, B. Wang, Y. Fang, *Int. J. Heat and Mass Transfer* **2024**, *220*, 125001.
[13] H. Lu, M. Zhang, L. Yang, B. Hou, R. P. Martinez, M. Mi, J. Du, L. Deng, M. Wu, S. Chowdhury, X. Ma, Y. Hao, *Fundam. Res.* **2025**, *5*, 315.
[14] Y. Zhong, J. Zhang, S. Wu, L. Jia, X. Yang, Y. Liu, Y. Zhang, Q. Sun, *Fundam. Res.* **2022**, *2*, 462.
[15] G. Sankar, A. Chandrasekhara Rao, P. S. Seshadri, K. R. Balasubramanian, *Appl. Therm. Eng.* **2016**, *103*, 1470.
[16] J. He, J. Wei, S. Yang, Y. Wang, K. Zhong, K. J. Chen, *IEEE Trans. Electron Devices* **2019**, *66*, 3453.
[17] K. Jagannadham, *Solid-State Electron.* **1998**, *42*, 2199.
[18] R. Soman, M. Malakoutian, B. Shankar, D. Field, E. Akso, N. Hatui, N. J. Hines, S. Graham, U. K. Mishra, M. Kuball, S. Chowdhury, *Tech. Dig.–Int. Electron Devices Meet.* **2022**, 3081.
[19] M. Trejo, G. H. Jessen, K. D. Chabak, J. K. Gillespie, A. Crespo, M. Kossler, V. Trimble, D. Langley, E. R. Heller, B. Claflin, D. E. Walker, B. Poling, R. Gilbert, G. D. Via, J. Hoelscher, J. Roussos, F. Ejeckam, J. Zimmer, *Phys. Status Solidi A* **2011**, *208*, 439.
[20] A. Siddique, R. Ahmed, J. Anderson, M. Nazari, L. Yates, S. Graham, M. Holtz, E. L. Piner, *ACS Appl. Electron. Mater.* **2019**, *1*, 1387.
[21] B. L. Hancock, M. Nazari, J. Anderson, E. Piner, F. Faili, S. Oh, D. Twitchen, S. Graham, M. Holtz, *Appl. Phys. Lett.* **2016**, *108*, 211901.
[22] Y. Zhou, R. Ramaneti, J. Anaya, S. Korneychuk, J. Derluyn, H. Sun, J. Pomeroy, J. Verbeeck, K. Haenen, M. Kuball, *Appl. Phys. Lett.* **2017**, *111*, 041901.
[23] S. Mandal, C. Yuan, F. Massabuau, J. W. Pomeroy, J. Cuenca, H. Bland, E. Thomas, D. Wallis, T. Batten, D. Morgan, R. Oliver, M. Kuball, O. A. Williams, *ACS Appl. Mater. Interfaces* **2019**, *11*, 40826.
[24] K. Wang, K. Ruan, W. Hu, S. Wu, H. Wang, *Scr. Mater.* **2020**, *174*, 87.
[25] A. Kobayashi, H. Tomiyama, Y. Ohno, Y. Shimizu, Y. Nagai, N. Shigekawa, J. Liang, *Funct. Diamond* **2022**, *2*, 142.
[26] T. Matsumae, Y. Kurashima, H. Takagi, Y. Shirayanagi, S. Hiza, K. Nishimura, E. Higurashi, *Scr. Mater.* **2022**, *215*, 114725.
[27] D. Francis, F. Faili, D. Babić, F. Ejeckam, A. Nurmikko, H. Maris, *Diamond Relat. Mater.* **2010**, *19*, 229.
[28] J. C. Mendes, M. Liehr, C. Li, *Materials* **2022**, *15*, 415.
[29] J. Liang, A. Kobayashi, Y. Shimizu, Y. Ohno, S. W. Kim, K. Koyama, M. Kasu, Y. Nagai, N. Shigekawa, *Adv. Mater.* **2021**, *33*, 2104564.

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[30] R. Kagawa, Z. Cheng, K. Kawamura, Y. Ohno, C. Moriyama, Y. Sakaida, S. Ouchi, H. Uratani, K. Inoue, Y. Nagai, N. Shigekawa, J. Liang, *Small* **2023**, *20*, 2305574.
[31] R. Kagawa, K. Kawamura, Y. Sakaida, S. Ouchi, H. Uratani, Y. Shimizu, Y. Ohno, Y. Nagai, J. Liang, N. Shigekawa, *Appl. Phys. Express* **2022**, *15*, 041003.
[32] J. C. Arnault, S. Saada, V. Ralchenko, *Phys. Status Solidi RRL* **2022**, *16*, 2100354.
[33] S. W. Kim, R. Takaya, S. Hirano, M. Kasu, *Appl. Phys. Express* **2021**, *14*, 115501.
[34] G. Aiello, S. Schreck, K. A. Avramidis, T. Franke, G. Gantenbein, J. Jelonnek, A. Meier, T. Scherer, D. Strauss, M. Thumm, M. Q. Tran, C. Wild, E. Woerner, *Fusion Eng. Des.* **2020**, *157*, 111818.
[35] B. S. El-Dasher, J. J. Gray, J. W. Tringe, J. Biener, A. V. Hamza, C. Wild, E. Wörner, P. Koidl, *Appl. Phys. Lett.* **2006**, *88*, 241915.
[36] C. Xiao, F. C. Hsia, A. Sutton-Cook, B. Weber, S. Franklin, *Carbon* **2022**, *196*, 29.
[37] H. Zhang, Z. Yan, Z. Song, S. Zhou, Z. Zhang, G. Chen, M. Liao, G. Chen, *Funct. Diamond* **2023**, *3*, 2246495.
[38] T. Miyatake, K. Kawaguchi, M. Ohtomo, T. Iwai, T. Ishiguro, Y. Doi, J. Hermias, S. Nur, R. Ishihara, S. Sato, *Jpn. J. Appl. Phys.* **2023**, *62*, 096503.
[39] J. Liang, Y. Nakamura, Y. Ohno, Y. Shimizu, Y. Nagai, H. Wang, N. Shigekawa, *Funct. Diamond* **2021**, *1*, 110.
[40] R. Kagawa, Y. Ohno, Y. Nagai, N. Shigekawa, J. Liang, *Funct. Diamond* **2024**, *4*, 2337352.
[41] K. Kawamura, S. Hishiki, H. Uratani, S. Ouchi, H. Suzuki, Y. Sakaida, K. Kitahara, H. Goto, I. Hide, S. Kaneko, O. Machida, T. Yoshie, *Int. Conf. Solid State Devices Mater.* **2020**, 283.
[42] Z. Cheng, J. Liang, K. Kawamura, H. Asamura, H. Uratani, S. Graham, Y. Ohno, Y. Nagai, N. Shigekawa, D. G. Cahill, *Nat. Commun.* **2022**, *13*, 7201.
[43] J. Liang, Y. Ohno, Y. Yamashita, Y. Shimizu, S. Kanda, N. Kamiuchi, S. Kim, K. Koji, Y. Nagai, M. Kasu, N. Shigekawa, *ACS Appl. Nano Mater.* **2020**, *3*, 2455.
[44] B. Monemar, *Phys. Rev. B* **1974**, *10*, 676.
[45] Y. P. Varshni, *Physica* **1967**, *34*, 149.
[46] R. B. Simon, J. Anaya, F. Faili, R. Balmer, G. T. Williams, D. J. Twitchen, D. M. Kuball, *Appl. Phys. Express* **2016**, *9*, 061302.
[47] N. A. Katcho, J. Carrete, W. Li, N. Mingo, *Phys. Rev. B* **2014**, *90*, 094117.
[48] D. Spiteri, J. Anaya, M. Kuball, *J. Appl. Phys.* **2016**, *119*, 085102.
[49] Z. Cheng, T. Bougher, T. Bai, S. Y. Wang, C. Li, L. Yates, B. M. Foley, M. Goorsky, B. A. Cola, F. Faili, S. Graham, *ACS Appl. Mater. Interfaces* **2018**, *10*, 2400387.
[50] Y. Yamamoto, T. Imai, K. Tanabe, T. Tsuno, Y. Kumazawa, N. Fujimori, *Diamond Relat. Mater.* **1997**, *6*, 1057.
[51] Z. Cheng, L. Yates, J. Shi, M. J. Tadjer, K. D. Hobart, S. Graham, *APL Mater* **2019**, *7*, 031118.
[52] A. V. Sukhadolau, E. V. Ivakin, V. G. Ralchenko, A. V. Khomich, A. V. Vlasov, A. F. Popovich, *Diamond Relat. Mater.* **2005**, *14*, 589.
[53] J. Liang, S. Nishida, M. Arai, N. Shigekawa, *J. Appl. Phys.* **2016**, *120*, 034504.
[54] J. Liang, T. Miyazaki, M. Morimoto, S. Nishida, N. Shigekawa, *J. Appl. Phys.* **2013**, *114*, 183703.
[55] E. Higurashi, K. Okumura, K. Nakasuji, T. Suga, *Jpn. J. Appl. Phys.* **2015**, *54*, 030207.
[56] J. Liang, D. Takatsuki, M. Higashiwaki, Y. Shimizu, Y. Ohno, Y. Nagai, N. Shigekawa, *Jpn. J. Appl. Phys.* **2022**, *61*, SF1001.
[57] J. Liang, S. Masuya, S. Kim, T. Oishi, M. Kasu, N. Shigekawa, *Appl. Phys. Express* **2019**, *12*, 016501.
[58] X. Chen, S. Boumaiza, L. Wei, *IEEE Trans. Electron Devices* **2019**, *66*, 3748.
[59] D. C. Dumka, T. M. Chou, J. L. Jimenez, D. M. Fanning, D. Francis, F. Faili, F. Ejeckam, M. Bernardoni, J. W. Pomeroy, M. Kuball, presented at *IEEE Compd. Semicond. Integr. Circuit Symp*, Monterey, CA, USA, October, **2013**.
[60] B. K. Schwitter, A. E. Parker, S. J. Mahon, A. P. Fattorini, M. C. Heimlich, *IEEE Trans. Electron Devices* **2014**, *61*, 1327.
[61] B. M. Paine, T. Rust, E. A. Moore, *IEEE Trans. Electron Devices* **2016**, *63*, 590.
[62] X. Ji, Z. Huang, Y. Ohno, K. Inoue, Y. Nagai, Y. Sakaida, H. Uratani, J. Sun, N. Shigekawa, J. Liang, Z. Cheng, *Adv. Electron. Mater.* **2024**, *11*, 2400387.
[63] M. Malakoutian, K. Woo, D. Rich, R. Mandia, X. Zheng, A. Kasperovich, D. Saraswat, R. Soman, Y. Jo, T. Pfeifer, T. Hwang, H. Aller, J. Kim, J. Lyu, J. K. Mabrey, T. A. Rodriguez, J. Pomeroy, P. E. Hopkins, S. Graham, D. J. Smith, S. Mitra, K. Cho, M. Kuball, S. Chowdhury, *Adv. Electron. Mater.* **2024**, *11*, 2400146.
[64] M. Malakoutian, D. E. Field, N. J. Hines, S. Pasayat, S. Graham, M. Kuball, S. Chowdhury, *ACS Appl. Mater. Interfaces* **2021**, *13*, 60553.

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